

Fixed

Helix OTA — Fixed (closed workable items)

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This is the canonical closed-archive tracker (§11.4.19 column alignment, §11.4.33 type-aware closure vocabulary, §11.4.54 OTA-NNN). Open items live in [Issues.md](#); the short-form companion is [Fixed_Summary.md](#). Sorted closure-date DESC. All commit hashes below were read from `git log`, not invented.

§3. [OTA-003] Emulator Tier-2 — Android emulator OTA lifecycle validation

Status: Completed (→ Fixed.md) **Type:** Task

Closed: 2026-06-21 **Root cause:** Android emulator CZ_API36_Phone (Android 16 API 36) was running on nezha.local but had no validated OTA lifecycle against the control plane — the register → update-check → telemetry flow was unproven. **Fix:** Booted emulator through containers submodule wrapper (`scripts/boot_android_emulator.sh`). Cross-compiled and deployed ota-server for linux/amd64. Full OTA lifecycle validated: admin login, device registration (emu64xa-AABBCCDD), device listing, client update check, telemetry reporting. All 7/7 tests PASS. **Evidence:** `docs/qa/ota-003-validation/STATUS.md`, `docs/qa/ota-003-validation/test5-ota-api-cycle.txt`.

§5. [OTA-021] HelixTrack bidirectional sync verification

Status: Completed (→ Fixed.md) **Type:** Task

Closed: 2026-06-21 **Root cause:** HelixTrack push and pull scripts had no formal end-to-end verification — sync worked but was unproven with captured evidence. **Fix:** Created and executed end-to-end sync test (11/11 PASS) covering both directions: push 5 items from `workable_items.db` to HelixTrack API with correct status/type mapping, pull tickets from API and idempotently update-or-create in the

local DB. Push/pull scripts (scripts/sync_helixtrack_push.sh, scripts/sync_helixtrack_pull.sh) committed and verified. GitFlic 100 MB packfile-limit recovery documented via scripts/reassemble_gitflic_bundle.sh.

Evidence: qa-results/helixtrack/20260620T201949Z/sync_output.txt, scripts/sync_helixtrack_push.sh, scripts/sync_helixtrack_pull.sh, docs/helixtrack_sync_state.md.

§1. [OTA-001] Telemetry deployment_id not derivable from ota-protocol.UpdateAvailable

Status: Fixed (→ Fixed.md) **Type:** Bug

Closed 2026-06-10 — commit 3c57867

feat(api): close 2 protocol gaps – UpdateAvailable.deployment_id + GET /deployments. The ota-protocol UpdateAvailable payload now carries the deployment_id field, wired through the server's update-check response, so a real device can obtain and echo back the deployment_id its telemetry must supply (previously the server required it but the protocol never told the device). Proven by TestEmulatorSelfServesDeploymentID — the Tier-1 emulator round-trips register → update-check → telemetry self-serving the deployment_id end-to-end.

§2. [OTA-002] No GET /deployments list endpoint to enumerate deployments

Status: Implemented (→ Fixed.md) **Type:** Feature

Closed 2026-06-10 — commit 3c57867

feat(api): close 2 protocol gaps – UpdateAvailable.deployment_id + GET /deployments. A GET /deployments list endpoint now enumerates existing deployments with the same cursor-pagination convention as the group/members endpoints (50ef5c6), so the dashboard and operator/automation clients can discover deployment IDs without already knowing them. Proven by TestDeploymentListReturnsActive — the list endpoint returns the active deployments.

§5. [OTA-005] Per-device telemetry filters (?event / ?since / ?until)

Status: Implemented (→ Fixed.md) **Type:** Feature

Closed 2026-06-10 — commit 50ef5c6 feat(api): per-device telemetry filters + group/members pagination. Per-device telemetry endpoint now accepts ?event, ?since, ?until query filters; OpenAPI synced (redocly-clean).

§6. [OTA-006] Group + group-members cursor pagination

Status: Implemented (→ Fixed.md) **Type:** Feature

Closed 2026-06-10 — commit 50ef5c6 feat(api): per-device telemetry filters + group/members pagination. Group and group-members listing endpoints gained cursor-based pagination.

§7. [OTA-007] Dashboard API client synced to new pagination/filter params

Status: Completed (→ Fixed.md) **Type:** Task

Closed 2026-06-10 — commit b0b8ee2 chore(dashboard): sync API client to new pagination/filter params. Dashboard API client brought into lockstep with the new telemetry-filter + pagination parameters (OTA-005/OTA-006).

§8. [OTA-008] Tier-1 Go OTA device-emulator

Status: Implemented (→ Fixed.md) **Type:** Feature

Closed 2026-06-10 — commit 7dc3334 feat(emulator): Tier-1 Go OTA device-emulator + resilience for new handlers. A Go device-emulator (server/internal/deviceemu + cmd/ota-device-emu) that speaks the real ota-protocol to the control plane; surfaced the telemetry deployment_id protocol gap now tracked as OTA-001.

§9. [OTA-009] Comprehensive dashboard UI testing system

Status: Completed (→ Fixed.md) **Type:** Task

Closed 2026-06-10 — commit fa571b8 test(dashboard): comprehensive UI testing system for all panels. UI test suite covering all dashboard panels (Vitest + Playwright + ally).

§10. [OTA-010] Autonomous e2e + security + HelixQA for telemetry filters & pagination

Status: Completed (→ Fixed.md) **Type:** Task

Closed 2026-06-10 — commit a839220 test(qa): autonomous e2e + security + HelixQA for telemetry filters & pagination. Autonomous end-to-end + security suites + HelixQA bank covering the OTA-005/OTA-006 telemetry-filter and pagination work.

§11. [OTA-015] A/B slot switch via U-Boot BOOT_ORDER

Status: Implemented (→ Fixed.md) **Type:** Feature

Closed 2026-06-11 — commit 18ed84a
feat(AB): PWU-AB-1 A/B slot switch proven. Full A/B slot switch via U-Boot BOOT_ORDER env var against real U-Boot 2024.01 + QEMU virt + HVF on the emulator T1 ladder. Boots A → reboots to B via fw_setenv → verifies HELIX_SLOTID=B, HELIX_ROOTDEV=/dev/vda3. Proven 3/3 deterministic. Evidence: docs/qa/20260611T094958Z-ab-slot-switch/. Runtime signature registered in docs/design/rk3588_ab_virt/runtime-signatures.yaml (PWU-AB-1-SLOT-SWITCH).

§12. [OTA-016] RAUC dd-apply to inactive slot with dm-verity

Status: Implemented (→ Fixed.md) **Type:** Feature

Closed 2026-06-20 — PWU-AB-2 RAUC reconciliation: dd rootfs to inactive slot + fw_setenv BOOT_ORDER switch so guest boots the new slot. RAUC slot-class scheme reconciled with uboot_ab/boot.cmd BOOT_ORDER env scheme. Proven 3/3 deterministic. Evidence: docs/qa/20260620T051026Z-ab-rauc-verity/. Runtime signature: PWU-AB-2-RAUC-DD.

§13. [OTA-017] U-Boot corrupt-slot auto-rollback via bootcount

Status: Implemented (→ Fixed.md) **Type:** Feature

Closed 2026-06-11 — commit 42be557 feat(AB): PWU-AB-3 corrupt-slot AUTO-ROLLBACK proven. bootcount > bootlimit triggers altbootcmd which swaps to the good slot. ROLLBACK run booted from bad slot B and rolled back to good slot A (HELIX_SLOTID=A, HELIX_ROOTDEV=/dev/vda2). Proven 2/2 deterministic. Evidence: docs/qa/20260611T095918Z-ab-rollback/. Runtime signature: PWU-AB-3-ROLLBACK.

§14. [OTA-018] ApplyPort CLI + slot manager + Ed25519 verifier

Status: Implemented (→ Fixed.md) **Type:** Feature

Closed 2026-06-20 — PWU-AB-4 ApplyPort (server/cmd/applyport/). Slot detection from /proc/cmdline (helix_slot=A|B), Ed25519 artifact signature verification via crypto/ed25519 (proven real, not a stub — §1.1 mutation TestMutationSignatureUsesRealEd25519), write-and-arm with health marker, and device client (login, check-for-update, apply). 58/58 tests passing including §1.1 paired-mutation suite. Evidence: server/internal/device/. Runtime signatures: PWU-AB-4-APPLYPORT-BUILD, PWU-AB-4-APPLYPORT-TESTS, PWU-AB-4-SLOT-DETECTION, PWU-AB-4-SIGNATURE-VERIFIER.

§9. [OTA-019] Build resource stats tracker

Status: Completed (→ Fixed.md) **Type:** Task

Closed: 2026-06-20 — commit 300351c6 feat(stats): OTA-002 Build resource stats tracker. **Root cause:** §11.4.24 mandated per-build resource telemetry; missing entirely. **Fix:** scripts/resource_sampler.sh — background-capable host-side sampler (memory RSS, CPU%, load, disk I/O at 5s intervals, min/max/mean/p95). docs/Stats.tsv TSV registry. docs/Stats.md human-readable report. Tested end-to-end (10s run produces correct TSV + Stats.md). Observer constraint verified: <50MB RSS, <5% CPU. **Evidence:** docs/Stats.tsv, docs/Stats.md, tests/stats_resource_sampler_test.sh.

§15. [OTA-014] Docs Chain submodule distribution (§11.4.106 Phase 6)

Status: Completed (→ Fixed.md) **Type:** Task

Closed: 2026-06-20 **Root cause:** Docs Chain engine was used in-tree but not registered as a git submodule (§11.4.106 Phase 6). **Fix:** Registered vasic-digital/docs_chain@2c8cf16 as submodule. Added helix-deps.yaml entry. Fixed 2 context YAMLS (features-status fingerprint block, helixtrack inline transforms). All 4 contexts pass doctor. **Evidence:** .gitmodules, helix-deps.yaml, docs_chain doctor --all (4/4 PASS).

§16. [OTA-020] Database migration test

Status: Fixed (→ Fixed.md) **Type:** Bug

Closed: 2026-06-21 **Root cause:** workable_items.db had no automated validation for schema (tables, columns, constraints) or sync consistency with Issues.md/Fixed.md. **Fix:** Created scripts/testing/test_workable_items_db.sh — 47 tests covering schema validation, Issues sync (4/4), Fixed sync (12/12), orphan

detection, terminal status checks. All 47/47 PASS. **Evidence:** qa-results/
workable_items/20260620T201912Z/, scripts/testing/
test_workable_items_db.sh, scripts/testing/
migrate_workable_items.sh.